

Glycan Drawing Tool Document

Introduction

Glycan Drawing Tool is a web-based tool for drawing and editing glycan structures.

Users can easily create monosaccharide nodes, linkage relationships, and text annotations through a visual interface. The tool automatically generates standard symbolic representations according to the **SNFG (Symbol Nomenclature for Glycans)** standard.

The interaction logic is similar to **ChemDraw**, providing researchers with a more consistent user experience.

Overview

The interface consists of four main parts:

- **Left Panel:** Toolbar (Select, Add Sugar, Add Text, Add Preset Glycan, Eraser)
- **Center Canvas:** Main drawing area (supports zooming and panning)
- **Right Panel:** Detailed customization options for sugar shape, color, linkage style, and text style
- **Top Panel:** Canvas operations (Undo/Redo, Clear Canvas, Export, Canvas Zoom)

Recent Features

The current version adds the following tools and interface improvements:

- **Render Presets** --- Each sugar can use **Flat**, **Soft**, or **Glossy** rendering. Mixed selections show no pressed render button.
 - **Add-Mode Linkage Tab** --- Configure linkage text, line style, label style, and direction before creating a new connection.
 - **Rotation and Alignment** --- Use the compass dial, numeric angle input, or preset angle buttons. Alignment, branch nudging, and linkage-length normalization are also available in the View tab.
 - **PNG Clipboard Copy** --- **Ctrl+Alt+C** copies selected sugars, texts, and connections as a PNG image for pasting into Word, PowerPoint, or other applications.
 - **Add-Mode Double Click** --- Double-click an existing sugar while in Add Sugar mode to switch to Select mode and adjust that sugar.
 - **Tool Shortcuts** --- **Alt+A** Add Sugar, **Alt+T** Add Text, **Alt+S** Select, **Alt+P** Preset Glycans, **Alt+D** Delete.
 - **Visitor Statistics** --- The left panel shows Busuanzi page views and visitors, with labels that follow the active language.
 - **Font and Version Display** --- Chinese uses Noto Sans SC, English uses Novecento Sans with a Google font fallback. The current version number appears beside the title.
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Toolbar Description

Tool	Icon	Function
Select		Select existing elements to move, copy, or modify their styles
Add Sugar		Click on the canvas to place a new monosaccharide symbol
Add Text		Add text annotations or labels on the canvas
Preset Glycans		Insert a complete glycan structure from built-in templates
Eraser		Delete selected elements

Add Sugar Mode

Available actions:

- **Add Monosaccharide** --- In add-sugar mode, click on the canvas to add a sugar.
- **Directional Addition** --- Click or long-press an existing sugar and drag toward a blank area to add a new sugar at a fixed angle.
- **Add Linkage** --- Long-press an existing sugar and drag toward another sugar to create a linkage between them.
- **Free End** --- Allows users to represent optional free ends. The asterisk symbol will not be exported in the final image.
- **Preset Style** --- Sugar and linkage styles adjustable in select mode (described below) can also be preconfigured and directly added here.

Select Mode

Available actions:

- **Edit Element Style** --- Click or drag-select sugars, linkages, or texts to adjust or update size, shape, and other properties.
- **Move Elements** --- Drag selected sugars to change their positions.
- **Batch Selection** --- Drag to select multiple elements for batch adjustment, or use **Shift + click** for precise multi-selection.
- **Keyboard Shortcuts** ---
 - **Ctrl/Cmd + Z/Y**: Undo/Redo
 - **Ctrl/Cmd + B/I/U**: Bold/Italic/Underline for text
 - **Ctrl/Cmd + C/X/V/A**: Copy/Cut/Paste/Select All
 - **Delete / Backspace**: Delete selected elements
 - **Esc**: Exit the current mode (e.g., preset placement)

Sugar Structure Styles

1. SNFG Sugar Presets

The right panel provides commonly used monosaccharide presets (Glc, Gal, Man, GlcNAc, GalNAc, Fuc, GlcA, Neu5Ac, Xyl), following the **SNFG 2.0** standard. These presets have default style settings in add sugar mode.

- By default, for visual consistency, sugars with **diamond** symbols (GlcA, Neu5Ac) are **2 px larger**, and those with **square** symbols (GlcNAc, GalNAc) are **2 px smaller** than other sugars. However, applying preset styles in select mode will not alter sugar sizes and borders, only filling colors and shapes.
- Other standard monosaccharides can be combined using the predefined shapes and colors. For details, please refer to the SNFG link provided in the left panel.

2. Custom Sugar Types

Users can customize:

- **Shape** --- Circle, square, diamond, triangle, star, parenthesis, or free-end symbol.
 - The asterisk (*) in the last category will **not appear in exported images**, allowing display of unlinked free ends.
- **Fill Color and Transparency** --- Preset SNFG 2.0 standard colors are available, or users can specify custom RGB hex colors.
- **Size**
- **Outline** --- Width, style, color, transparency.



Linkage and Linkage Text

Customizable features:

- **Linkage Information** --- Enter full (e.g., α 2-6) or simplified (e.g., b14) linkage notation for each bond.
- **Linkage Display** --- Check the option to show linkage information on the right side of the bond. Orientation follows **SNFG 2.0** convention, with numbers closer to the free end. The "swap" button can reverse the linkage direction.
- **Link Line Style** --- Width, style, color, transparency.
- **Linkage Text Style** --- Font size, typeface, style, color, transparency.



Text Addition and Style

Using the "Add Text" tool, you can insert text labels anywhere on the canvas. Adjustable options include:

- Font size and typeface
- Bold / Italic / Underline
- Color and transparency



Preset Oligosaccharide Templates

Supports inserting complete glycan structure SVG templates. Click a preset thumbnail, then click on the canvas to place the template.



Eraser

Click or long-press to select and delete elements, or perform continuous deletion. **Delete / Backspace** shortcuts are supported.

Undo and Redo

Built-in full undo/redo system:

- **Undo** --- Revert the previous action
- **Redo** --- Reapply an undone action
- Saves up to **50 history steps**

Includes changes to style, position, addition, and deletion of elements.

Export and Canvas Control

Export Options

Format	Description
SVG	Editable vector format preserving all graphical information
PNG	High-resolution bitmap with transparent background
JPG	White-background bitmap suitable for presentation

Export Sizes

- Small (800×600)
- Medium (1000×700)
- Large (1200×800)

Canvas Operations

- **Pan:** Drag the canvas with the mouse
 - **Zoom:** Adjust zoom level (50%-300%) using the slider
 - **Reset:** Restore to 100% view
 - **Shortcut:** Alt + Scroll
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Technical Implementation

- **Frontend Framework:** Pure Vanilla JavaScript + SVG operations
 - **Main Class Structure:** `GlycanDrawer`
 - Manages canvas, event bindings, element drawing, and undo/redo system
 - **Core Logic Modules:**
 - `setupToolbar()` --- Toolbar interactions
 - `setupStyleControls()` --- Style control and binding
 - `initializeWorkspace()` --- Canvas initialization
 - `exportCanvas()` --- Export logic
 - `handleMouseDown/Move/Up()` --- User interaction handling
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Future Plans

- **Structure Conversion** --- Mutual conversion between standardized IUPAC glycan names and SNFG graphical representations
 - **Auto-Arrangement** --- One-click cleanup and alignment of glycan structures
 - **Position Adjustment** --- Horizontal/vertical movement / fixed bond-length movement
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Contact & Help

- **Developer:** x-yguo@pku.edu.cn
- **Reference Standard:** [NCBI SNFG 2.0](#)